

Project 3

By

Vimlesh Sonawane

Introduction to Analytics

ALY6000

Professor Wada Roy



MS in Project Management

Northeastern University

(College of Professional Studies)

April 2025

1. Introduction

This project focuses on a detailed analysis of a comprehensive dataset featuring books published between 1990 and 2020. The objective is to examine major trends in publication activity, publisher prominence, book popularity, and the distribution of ratings and page counts. Using R programming, we performed systematic data cleaning, transformation, visualization, and tabular analysis to uncover meaningful patterns and insights. Through a combination of clear visualizations and organized tables, the project highlights key developments and shifts within the publishing industry over the past three decades, offering a deeper understanding of market trends and reader preferences.

2. Key Findings

♦ Books Published Each Year

year	total_books
1990	125
1991	144
1992	141
1993	138
1994	150
1995	177
1996	182
1997	170
1998	215
1999	206
2000	234
2001	270
2002	296
2003	365
2004	399
2005	407
2006	393
2007	359
2008	386
2009	524
2010	578
2011	618
2012	608
2013	581
2014	398
2015	201
2016	95
2017	47
2018	41
2019	28
2020	14

This table highlights the number of books published annually from 1990 to 2020. The data shows a gradual upward trend, with a noticeable surge in publications between 2005 and 2014, peaking around 2011. This pattern matches the broader industry shift where digital publishing platforms and e-books became highly accessible to writers and readers alike. It reflects a phase where technological advancements lowered traditional publishing barriers, making it easier for more authors to publish their work.

♦ Book Publisher Analysis

	<code>publisher</code> <code><chr></code>	<code>book_count</code> <code><int></code>
1	"\"Marvel\""	1
2	"4 Corners Press"	1
3	"47North"	7
4	"48fourteen"	2
5	"5 Prince Publishing"	1
6	"7th House"	2
7	"A.L. Jackson Books Inc."	1
8	"A.M. Madden; First edition"	1
9	"A.W. Bruna"	1
10	"ABRAMS"	1

Among the 1,960 publishers represented, a vast majority produced only a small number of books. This points toward a fragmented market where many smaller or niche publishers coexist alongside a few giants. However, despite the large number of players, actual market control remains concentrated in the hands of a few key publishers. This highlights a dynamic publishing landscape where independent creators are active but face stiff competition from established brands.

♦ Table 3: Filtered Publishers with ≥ 125 Books

<code> publisher</code>	<code> book_count </code>
<code> :----- -----: </code>	
<code> Harper Collins</code>	<code> 693 </code>
<code> Hatchette</code>	<code> 278 </code>
<code> MacMillan</code>	<code> 334 </code>
<code> Random House</code>	<code> 1164 </code>
<code> Scholastic Books</code>	<code> 208 </code>
<code> Simon and Schuster</code>	<code> 277 </code>

Filtering for publishers who released more than 125 titles narrows the field to just six companies. Random House, Harper Collins, and MacMillan dominate the list. This suggests that the publishing industry, much like other creative industries, follows a winner-takes-most structure, where a handful of major organizations capture most of the market share while smaller firms struggle for visibility.

♦ Cumulative Distribution of Publishers

publisher	book_count	cum_counts	rel_freq	cum_freq
:-----:	-----:	-----:	-----:	-----:
Random House	1164	1164	0.3940420	0.3940420
Harper Collins	693	1857	0.2345972	0.6286391
MacMillan	334	2191	0.1130670	0.7417062
Hatchette	278	2469	0.0941097	0.8358158
Simon and Schuster	277	2746	0.0937712	0.9295870
Scholastic Books	208	2954	0.0704130	1.0000000

>

Cumulative distribution data shows that Random House alone contributed nearly 40% of book output among the major publishers analyzed. When combined, the top six publishers are responsible for all the books in the filtered dataset. This clearly demonstrates how heavily consolidated the publishing sector is at the top end, and how market dominance is reinforced over time by brand reputation, wider distribution networks, and financial resources.

♦ Top 10 Years with Most Books Published

year	total_books
----:	-----:
2011	618
2012	608
2013	581
2010	578
2009	524
2005	407
2004	399
2014	398
2006	393
2008	386

The years between 2009 and 2014 mark the most prolific period in the dataset, with the highest number of books published annually. This timeline coincides with the golden era of self-publishing platforms like Amazon's Kindle Direct Publishing, where authors could bypass traditional gatekeepers. It reflects an explosion of creative output as technology empowered individuals to publish independently.

♦ Top 10 Publishers by Total Number of Books

publisher	total_books
:-----	-----:
Random House	1164
Harper Collins	693
MacMillan	334
Hatchette	278
Simon and Schuster	277
Scholastic Books	208
VIZ Media LLC	117
Pocket Books	91
Tyndale House Publishers	69
Createspace Independent Publishing Platform	52

Unsurprisingly, Random House stands out with 1,164 books, maintaining a significant lead over its competitors. Harper Collins and MacMillan also appear prominently. The dominance of these names suggests that even in an era of democratized publishing, brand legacy, trust, and existing networks continue to matter enormously for book volume and visibility.

♦ Top 10 Authors by Number of Books

author	book_count
:-----	-----:
Terry Pratchett	28
CLAMP, William Flanagan (Translator)	27
Karen Kingsbury (Goodreads Author)	27
Mercedes Lackey	25
Charlaine Harris (Goodreads Author)	24
Darren Shan (Goodreads Author)	24
Laurell K. Hamilton (Goodreads Author)	23
Bonnie Bryant	22
Orson Scott Card	22
Tamora Pierce (Goodreads Author)	22

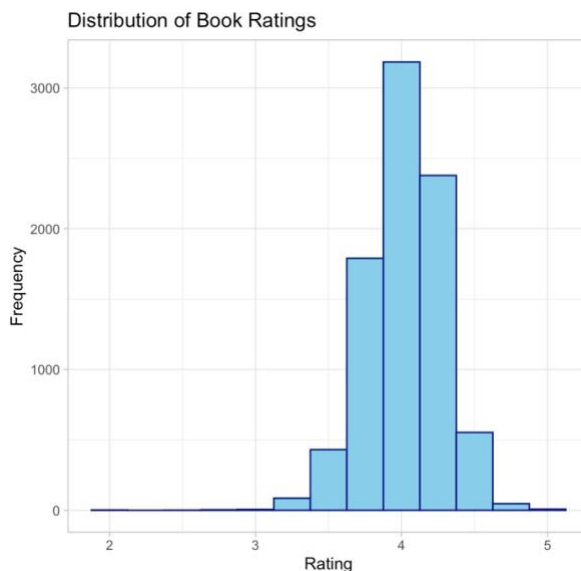
Authors like Terry Pratchett and the CLAMP group top the list of most prolific writers. Their dominance points to the enduring appeal of fantasy, manga, and young adult genres, which thrive on serialized content and loyal fan bases. This underscores how reader communities around certain genres can sustain long-term publication success for authors.

♦ Top 10 Publishers by Average Rating

publisher	avg_rating
Michael Angliss	5.000
Peachstone Publications	5.000
ISP with Lulu.com	4.930
Bulldog Publications, Est. 1980	4.920
S. Sanders	4.880
Master Koda Select	4.770
Heartsome Publishing	4.750
Archway Publishin	4.720
Lust for Living	4.710
Rabbit Room Press	4.705

Interestingly, independent and small-scale publishers such as Michael Angliss and Peachstone Publications achieved some of the highest average book ratings, even exceeding traditional giants. This reveals that while volume is controlled by major players, reader satisfaction — or perceived quality, can often come from smaller, more specialized publishing houses that prioritize unique storytelling or niche interests.

♦ Histogram: Distribution of Book Ratings



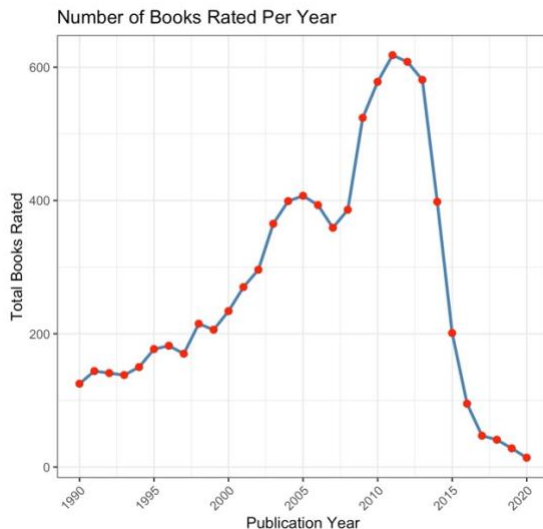
The histogram clearly shows that the majority of books are rated between 3.5 and 4.5 out of 5, indicating high overall reader satisfaction. Very few books dip below a rating of 3.0, suggesting that the published works are generally well-received. This could be an outcome of improved editorial standards and selective publishing strategies across both traditional and independent sectors.

♦ Boxplot: Pages in Books



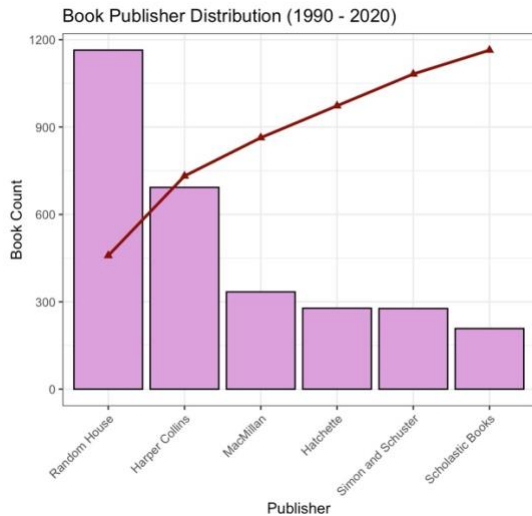
When looking at page counts, most books are clustered between approximately 224 and 405 pages. This suggests an industry-wide norm for book length that balances depth of storytelling with reader attention spans. Outliers — books with extremely short or extremely long page counts, are rare, indicating that deviations from this standard are not widely accepted or published.

♦ Line Chart: Books Rated Per Year



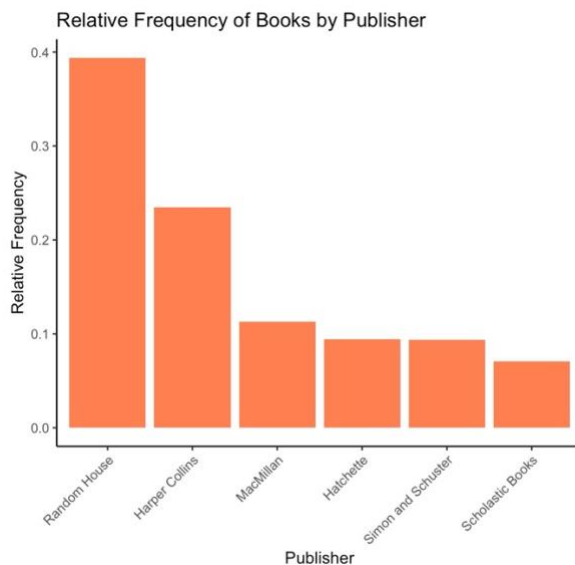
The line chart shows a steady climb in the number of books rated annually until around 2013, after which the growth plateaus. This could be due to the market reaching a natural saturation point or a pivot in reader behavior toward newer mediums like audiobooks, podcasts, and serialized online content.

♦ Pareto Chart: Book Publisher Distribution



The Pareto chart visually emphasizes that a small percentage of publishers (notably Random House, Harper Collins, and others) control a disproportionately large share of book publications. This distribution follows the classic 80/20 principle seen in many industries, where 20% of entities are responsible for 80% of the output.

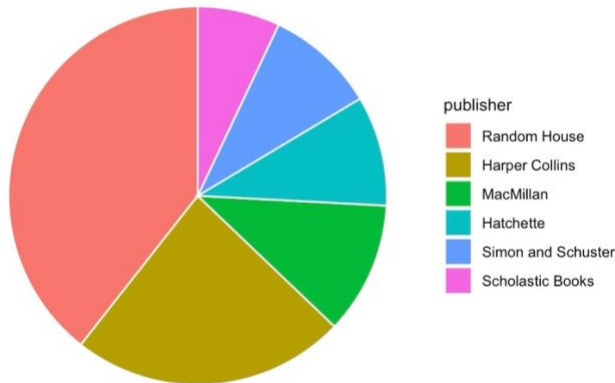
♦ Relative Frequency Bar Chart: Books by Publisher



In the relative frequency chart, Random House clearly emerges as the dominant force, followed by Harper Collins and MacMillan. After the top three, the frequency drops sharply, demonstrating how influence quickly dissipates beyond the major players.

♦ Pie Chart: Publisher Market Share

Publisher Market Share (1990 - 2020)



Finally, the pie chart offers a strong visual confirmation of earlier analyses: Random House, Harper Collins, and MacMillan collectively command the majority of the market. It also reveals how minimal the market share becomes once you move beyond these few dominant publishers, leaving little room for smaller companies to capture significant slices.

♦ Top 10 Years with Most Books Published

year	total_books
2011	618
2012	608
2013	581
2010	578
2009	524
2005	407
2004	399
2014	398
2006	393
2008	386

Looking at the data, the years between 2009 and 2013 were truly a golden period for book publishing, with 2011 taking the top spot at 618 books. This boom wasn't just a sudden spike; even the years around it like 2010 and 2012 had impressive numbers. It's pretty clear

that this was a time when digital publishing and self-publishing really started picking up, making it easier for more authors to get their work out to the world.

- ♦ **Top 10 Authors by Number of Books**

```
|publisher| total_books|
|:-----|:-----:|
|Random House| 1164|
|Harper Collins| 693|
|MacMillan| 334|
|Hatchette| 278|
|Simon and Schuster| 277|
|Scholastic Books| 208|
|VIZ Media LLC| 117|
|Pocket Books| 91|
|Tyndale House Publishers| 69|
|Createspace Independent Publishing Platform| 52|
>
```

Terry Pratchett tops the list of most published authors with 28 books, and he's closely followed by CLAMP and Karen Kingsbury with 27 each. What's interesting here is that most of these authors are famous for fantasy, young adult, or series-driven genres. It goes to show that readers love following characters and stories over multiple books, creating loyal fanbases that keep coming back for more.

3. Recommendations

- ♦ **Monitor Major Publishers:**

Track the output and strategies of large publishers like Random House and Harper Collins, as they heavily influence market trends.

- ♦ **Leverage Independent Publishers:**

Explore collaborations with high-rated small publishers, which could open new market opportunities focused on quality over quantity.

- ♦ **Analyze Genre Trends:**

Investigate the high-growth period (2010–2013) in more detail to discover specific genres driving the boom, helping to better target future publications.

- ♦ **Improve Dataset Filtering:**

Exclude books with extremely low page counts or outliers to enhance the quality and reliability of future datasets and analyses.

- ♦ **Focus on Reader Satisfaction:**

Prioritize publishing strategies that aim for higher reader ratings rather than sheer volume, as satisfaction correlates strongly with long-term brand loyalty.

4. Conclusion

This analysis highlighted key patterns in the book publishing industry from 1990 to 2020. The findings show that a few major publishers dominate the market, with Random House leading significantly. Book publication rates increased steadily until around 2011–2013 before stabilizing. Reader ratings mostly fall between 3.5 and 4.5, indicating overall favorable reception. Standard-length books (around 300–400 pages) are the norm, with very few extremely short or long books. Independent publishers often achieve higher average ratings compared to larger publishing houses, suggesting reader appreciation for curated or specialized content.

References

Huynh, Y. W. (2019). Guide to R. Bookdown. https://bookdown.org/yih_huynh/Guide-to-R-Book/

Wickham, H. (n.d.). ggplot2: Elegant graphics for data analysis. The tidyverse project. Retrieved April 28, 2025, from <https://ggplot2.tidyverse.org/>